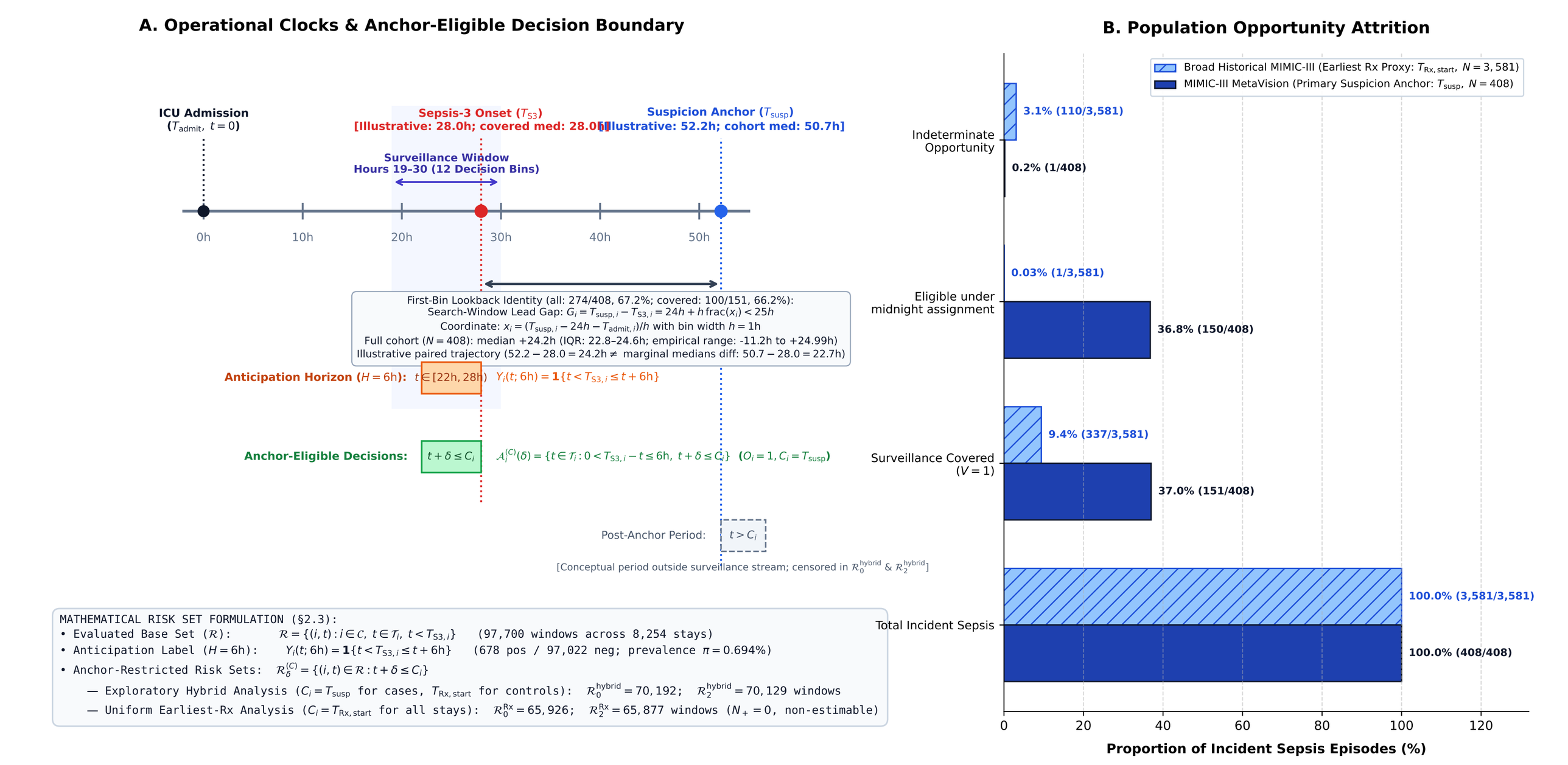

A. Operational Clocks & Anchor-Eligible Decision Boundary
ICU Admission
(T_{admit}, $t = 0$)
Sepsis-3 Onset (T_{S3})
[Illustrative: 28.0h; covered med: 28.0h]
Suspicion Anchor (T_{susp})
[Illustrative: 52.2h; cohort med: 50.7h]
Surveillance Window
Hours 19–30 (12 Decision Bins)
0h
10h
20h
30h
40h
50h
First-Bin Lookback Identity (all: 274/408, 67.2%; covered: 100/151, 66.2%):
Search-Window Lead Gap: $G_i = T_{susp,i} - T_{S3,i} = 24h + h\,\mathrm{frac}(x_i) < 25h$
Coordinate: $x_i = (T_{susp,i} - 24h - T_{admit,i})/h$ with bin width $h = 1$h
Full cohort ($N = 408$): median +24.2h (IQR: 22.8–24.6h; empirical range: -11.2h to +24.99h)
Illustrative paired trajectory ($52.2 - 28.0 = 24.2$h $\neq$ marginal medians diff: $50.7 - 28.0 = 22.7$h)
Anticipation Horizon ($H = 6$h): $t \in [22h, 28h)$ $Y_i(t; 6h) = \mathbf{1}\{t < T_{S3,i} \le t + 6h\}$
Anchor-Eligible Decisions: $t + \delta \le C_i$ $\mathcal{A}_i^{(C)}(\delta) = \{t \in \mathcal{T}_i : 0 < T_{S3,i} - t \le 6h,\ t + \delta \le C_i\}$ ($O_i = 1, C_i = T_{susp}$)
Post-Anchor Period: $t > C_i$
[Conceptual period outside surveillance stream; censored in $\mathcal{R}_0^{hybrid}$ & $\mathcal{R}_2^{hybrid}$]
MATHEMATICAL RISK SET FORMULATION (§2.3):
• Evaluated Base Set ($\mathcal{R}$): $\mathcal{R} = \{(i,t) : i \in C,\ t \in \mathcal{T}_i,\ t < T_{S3,i}\}$ (97,700 windows across 8,254 stays)
• Anticipation Label ($H = 6$h): $Y_i(t; 6h) = \mathbf{1}\{t < T_{S3,i} \le t + 6h\}$ (678 pos / 97,022 neg; prevalence $\pi = 0.694\%$)
• Anchor-Restricted Risk Sets: $\mathcal{R}_\delta^{(C)} = \{(i,t) \in \mathcal{R} : t + \delta \le C_i\}$
— Exploratory Hybrid Analysis ($C_i = T_{susp}$ for cases, $T_{Rx,start}$ for controls): $\mathcal{R}_0^{hybrid} = 70,192$; $\mathcal{R}_2^{hybrid} = 70,129$ windows
— Uniform Earliest-Rx Analysis ($C_i = T_{Rx,start}$ for all stays): $\mathcal{R}_0^{Rx} = 65,926$; $\mathcal{R}_2^{Rx} = 65,877$ windows ($N_+ = 0$, non-estimable)
B. Population Opportunity Attrition
Broad Historical MIMIC-III (Earliest Rx Proxy: $T_{Rx,start}$, $N = 3,581$)
MIMIC-III MetaVision (Primary Suspicion Anchor: T_{susp}, $N = 408$)
Indeterminate Opportunity
3.1% (110/3,581)
0.2% (1/408)
Eligible under midnight assignment
0.03% (1/3,581)
36.8% (150/408)
Surveillance Covered ($V = 1$)
9.4% (337/3,581)
37.0% (151/408)
Total Incident Sepsis
100.0% (3,581/3,581)
100.0% (408/408)
0
20
40
60
80
100
120
Proportion of Incident Sepsis Episodes (%)